

Assignment 5: Data Visualization

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Visualization

Directions

1. Rename this file `<FirstLast>_A05_DataVisualization.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up your session

1. Set up your session. Load the tidyverse, here & cowplot packages, and verify your home directory. Read in the NTL-LTER processed data files for nutrients and chemistry/physics for Peter and Paul Lakes (use the tidy NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv version in the Processed_KEY folder) and the processed data file for the Niwot Ridge litter dataset (use the NEON_NIWO_Litter_mass_trap_Processed.csv version, again from the Processed_KEY folder).
2. Make sure R is reading dates as date format; if not change the format to date.

#1

```
#Import basic libraries  
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
## v dplyr      1.1.4      v readr      2.1.5  
## v forcats    1.0.0      v stringr    1.5.1  
## v ggplot2    3.5.1      v tibble     3.2.1  
## v lubridate  1.9.3      v tidyr      1.3.1  
## v purrr      1.0.2  
## -- Conflicts ----- tidyverse_conflicts() --  
## x dplyr::filter() masks stats::filter()  
## x dplyr::lag()     masks stats::lag()  
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(here)
```

```
## here() starts at /home/guest/EDE_Fall2025
```

```
library(lubridate)
library(ggribes)
library(cowplot)
```

```
##
## Attaching package: 'cowplot'
##
## The following object is masked from 'package:lubridate':
##
##     stamp
```

```
library(viridis)
```

```
## Loading required package: viridisLite
```

```
library(RColorBrewer)
library(colormap)
library(ggthemes)
```

```
##
## Attaching package: 'ggthemes'
##
## The following object is masked from 'package:cowplot':
##
##     theme_map
```

```
library(ggplot2)
```

```
getwd()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
processed_data = here("Data/Processed_KEY")
```

```
PeterPaul.chem.nutrients <- read.csv(
  here("Data/Processed_KEY", "NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv"),
  stringsAsFactors = TRUE
)
```

```
NEON_NIWO_Litter <-
  read.csv (here
```

```

("Data/Processed_KEY", "NEON_NIWO_Litter_mass_trap_Processed.csv"),
stringsAsFactors = T)

```

#2

```

PeterPaul.chem.nutrients$sampldate <- ymd(PeterPaul.chem.nutrients$sampldate)
NEON_NIWO_Litter$collectDate <- ymd(NEON_NIWO_Litter$collectDate)

```

Define your theme

3. Build a theme and set it as your default theme. Customize the look of at least two of the following:

- Plot background
- Plot title
- Axis labels
- Axis ticks/gridlines
- Legend

#3

```

mytheme <- theme_minimal(base_size = 14) +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5, size = 16, color = "navy"),
    axis.title.x = element_text(face = "italic", size = 12),
    axis.title.y = element_text(face = "italic", size = 12),
    axis.text = element_text(color = "black"),
    legend.position = "top"
  )
theme_set(mytheme)

```

Create graphs

For numbers 4-7, create ggplot graphs and adjust aesthetics to follow best practices for data visualization. Ensure your theme, color palettes, axes, and additional aesthetics are edited accordingly.

4. [NTL-LTER] Plot total phosphorus (tp_{ug}) by phosphate (po₄), with separate aesthetics for Peter and Paul lakes. Add line(s) of best fit using the `lm` method. Adjust your axes to hide extreme values (hint: change the limits using `xlim()` and/or `ylim()`).

#4

```

PeterPaul.chem.nutrients %>%
  ggplot(aes(
    x = tp_ug,
    y = po4,
    color = po4
  )) +
  geom_point(alpha = 0.6) +

```

```
geom_smooth(method = "lm", se = FALSE, color = "black") +
facet_wrap(facets = vars(lakename),
           nrow=1) +
scale_color_gradient(low = "blue", high = "red") +
labs(
  title = "Concentration of Phosphate in Peter
and Paul lake by Total Phosphorus",
  x = "Total phosphorus (TP, µg/L)",
  y = "Phosphate (PO4, µg/L)",
  color = "Concentration"
) +
xlim(0, 125) +
ylim(0, 125)
```

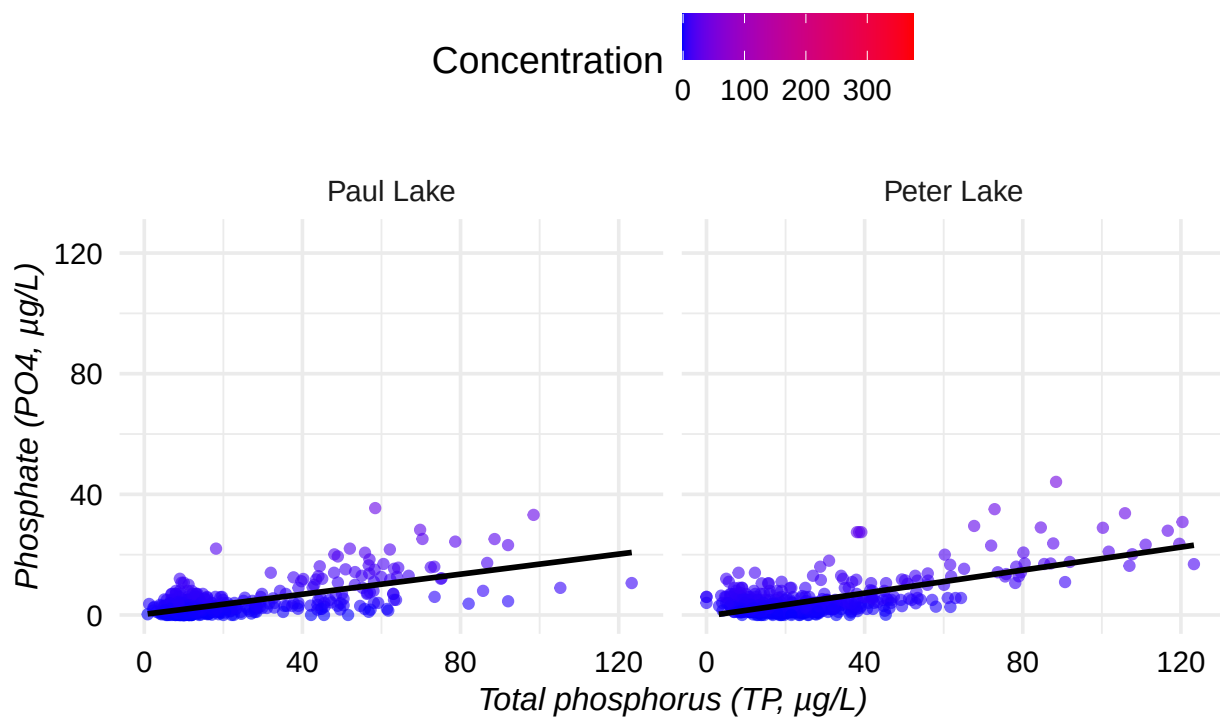
```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 21952 rows containing non-finite outside the scale range
## ('stat_smooth()').
```

```
## Warning: Removed 21952 rows containing missing values or values outside the scale range
## ('geom_point()').
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```

Concentration of Phosphate in Peter and Paul lake by Total Phosphorus



5. [NTL-LTER] Make three separate boxplots of (a) temperature, (b) TP, and (c) TN, with month as the x axis and lake as a color aesthetic. Then, create a cowplot that combines the three graphs. Make sure that only one legend is present and that graph axes are aligned.

Tips: * Recall the discussion on factors in the lab section as it may be helpful here. * Setting an axis title in your theme to `element_blank()` removes the axis title (useful when multiple, aligned plots use the same axis values) * Setting a legend's position to "none" will remove the legend from a plot. * Individual plots can have different sizes when combined using `cowplot`.

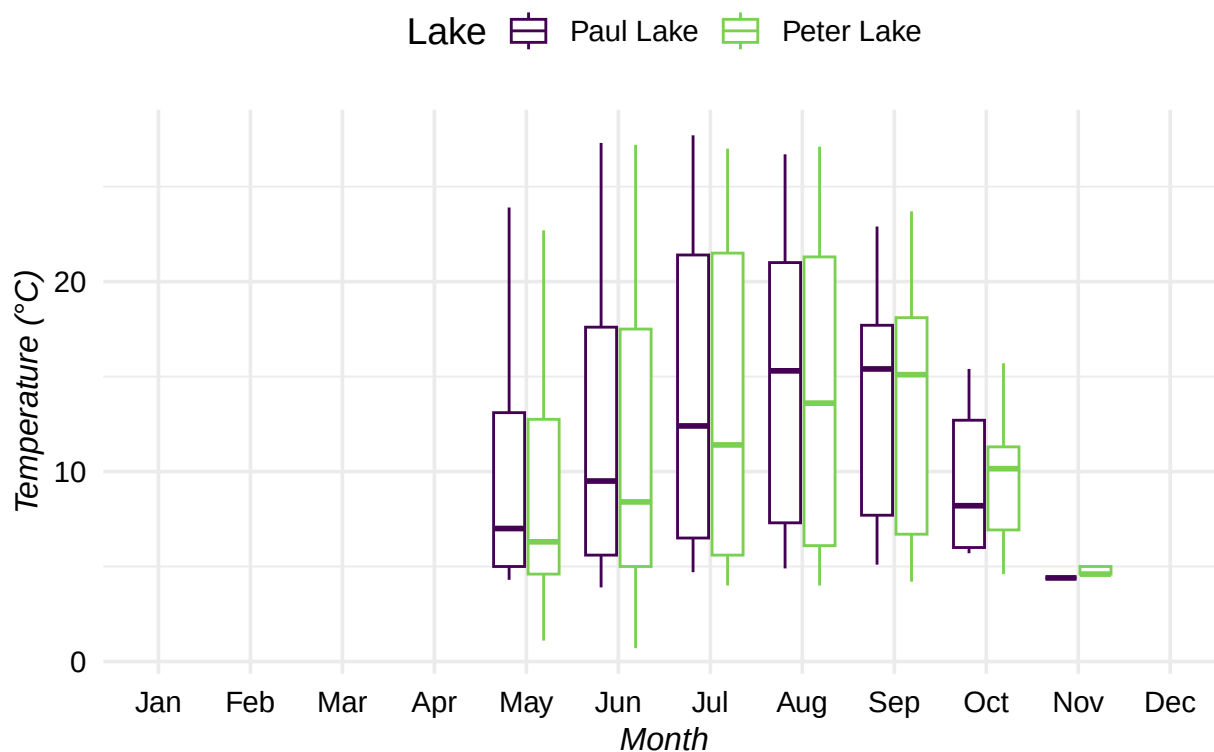
```
#5

#Temperature Plot

plot_temp <-ggplot(PeterPaul.chem.nutrients,
  aes(
    x = factor(month, levels = 1:12, labels = month.abb),
    y = temperature_C,
    color = lakename)
) +
  geom_boxplot()+
  scale_color_viridis(discrete = TRUE, end = 0.8)+
  scale_x_discrete(drop = FALSE) +
  labs(
    title = "Monthly Temperature by Lake",
    x = "Month",
    y = "Temperature (°C)",
    color = "Lake"
  ) +
  mytheme
print(plot_temp)
```

```
## Warning: Removed 3566 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

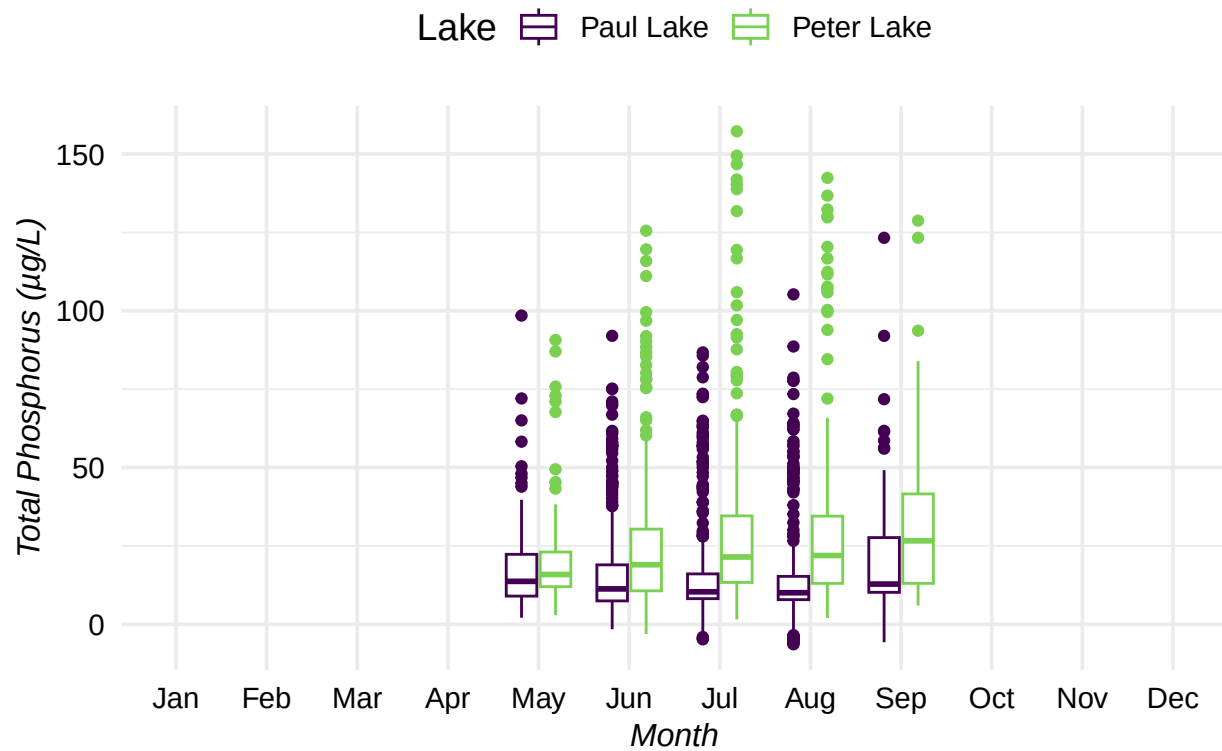
Monthly Temperature by Lake



```
#TP plot
plot_tp <-ggplot(
  PeterPaul.chem.nutrients,
  aes(
    x = factor(month, levels = 1:12, labels = month.abb),
    y = tp_ug,
    color = lakename)
) +
  geom_boxplot() +
  scale_color_viridis(discrete = TRUE, end = 0.8) +
  scale_x_discrete(drop = FALSE) +
  labs(
    title = "Monthly Phosphorus by Lake",
    x = "Month",
    y = "Total Phosphorus (µg/L)",
    color = "Lake"
  ) +
  mytheme
print(plot_tp)
```

```
## Warning: Removed 20729 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

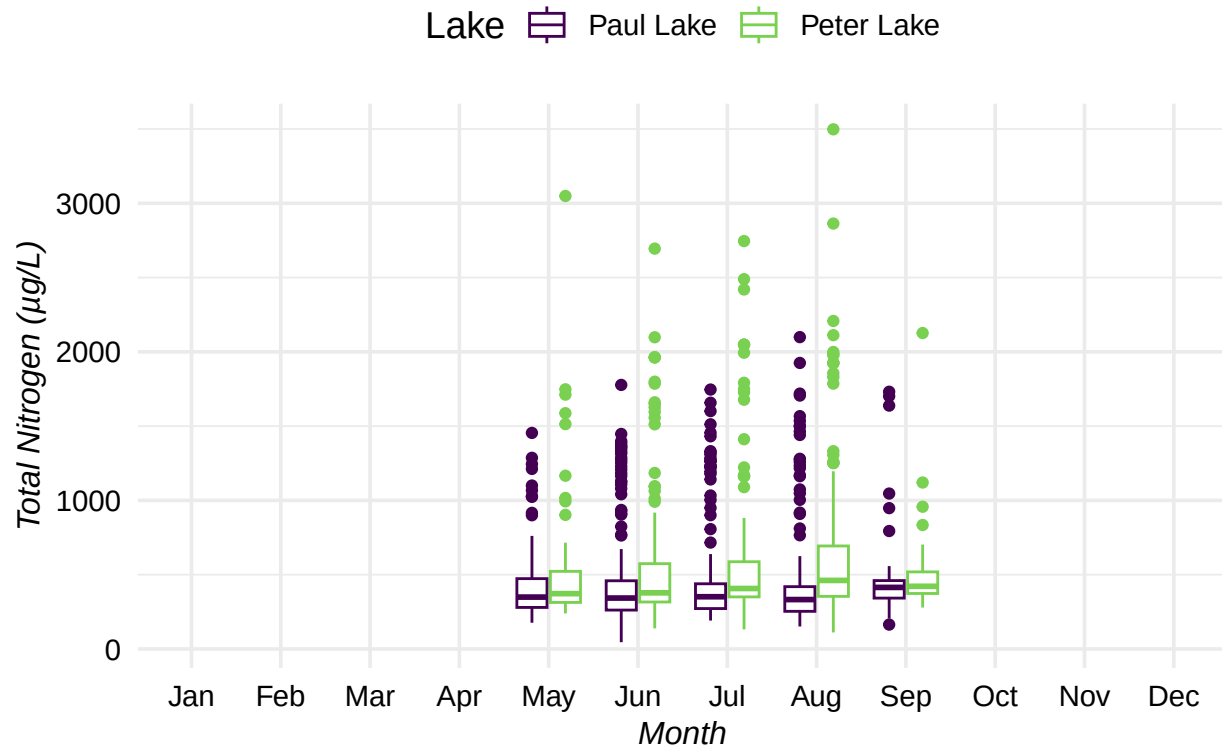
Monthly Phosphorus by Lake



```
#TN plot
plot_tn <- ggplot(
  PeterPaul.chem.nutrients,
  aes(
    x = factor(month, levels = 1:12, labels = month.abb),
    y = tn_ug,
    color = lakename)
) +
  geom_boxplot() +
  scale_color_viridis(discrete = TRUE, end = 0.8) +
  scale_x_discrete(drop = FALSE) +
  labs(
    title = "Monthly Total Nitrogen by Lake",
    x = "Month",
    y = "Total Nitrogen ( $\mu\text{g/L}$ )",
    color = "Lake"
  ) +
  theme
print(plot_tn)
```

```
## Warning: Removed 21583 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

Monthly Total Nitrogen by Lake



```
# Combined Couplot

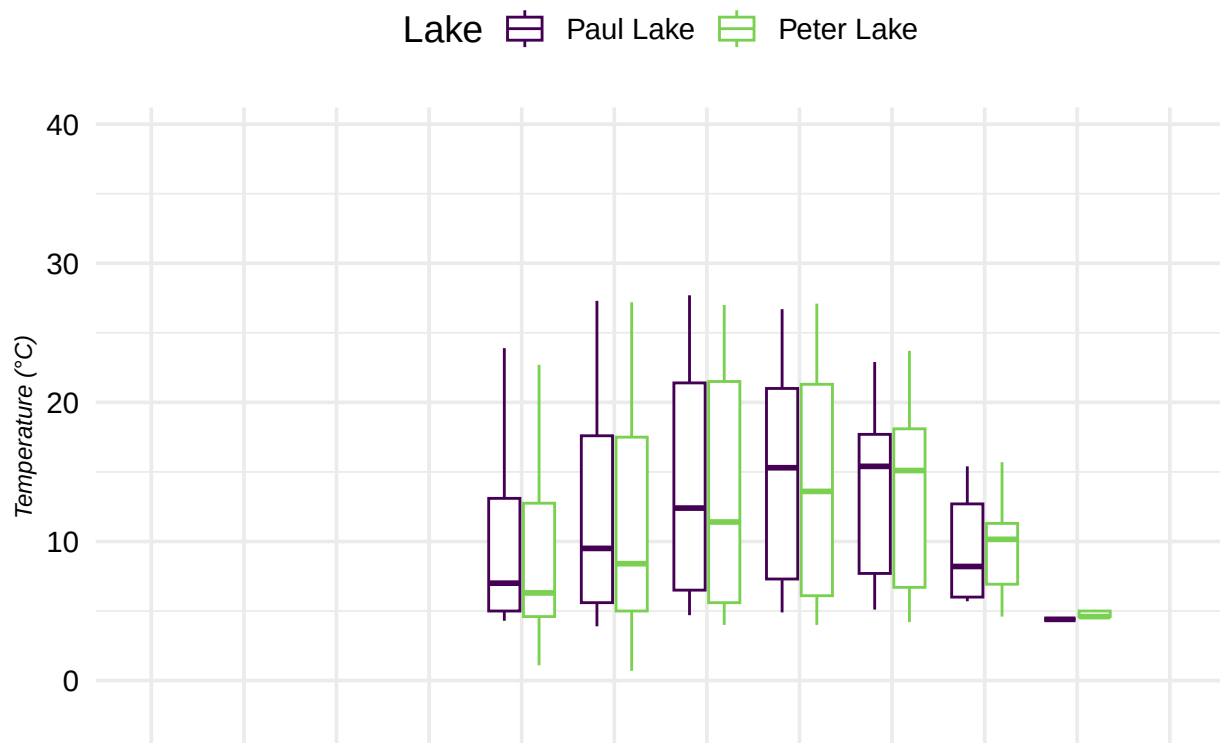
#Temp plot modified to keep legend and remove x-axis title and text

plot_temp_mod <-ggplot(PeterPaul.chem.nutrients,
  aes(
    x = factor(month, levels = 1:12, labels = month.abb),
    y = temperature_C,
    color = lakename)
) +
  geom_boxplot()+
  scale_color_viridis(discrete = TRUE, end = 0.8)+
  scale_x_discrete(drop = FALSE) +
  scale_y_continuous(expand = expansion(mult = c(0.2, 0.5)))+
  labs(
    title = "Monthly Temperature by Lake",
    x = "Month",
    y = "Temperature (°C)",
    color = "Lake"
  ) +
  mytheme +
  theme(
    axis.title.x = element_blank(),
    axis.text.x = element_blank(),
    axis.title.y = element_text(face = "italic", size = 9)
  )
```

```
print (plot_temp_mod)
```

```
## Warning: Removed 3566 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

Monthly Temperature by Lake



```
#TP plot modified to remove legend and remove x-axis title and text
plot_tp_mod <-ggplot(
  PeterPaul.chem.nutrients,
  aes(
    x = factor(month, levels = 1:12, labels = month.abb),
    y = tp_ug,
    color = lakename)
) +
  geom_boxplot() +
  scale_color_viridis(discrete = TRUE, end = 0.8) +
  scale_x_discrete(drop = FALSE) +
  scale_y_continuous(expand = expansion(mult = c(0.2, 0.5)))+
  labs(
    title = "Monthly Phosphorus by Lake",
    x = "Month",
    y = "Total Phosphorus (µg/L)",
    color = "Lake"
  ) +
  mytheme +
```

```

theme(
  legend.position = "none",
  axis.title.x = element_blank(),
  axis.text.x = element_blank(),
  axis.title.y = element_text(face = "italic", size = 9)
)
print(plot_tp_mod)

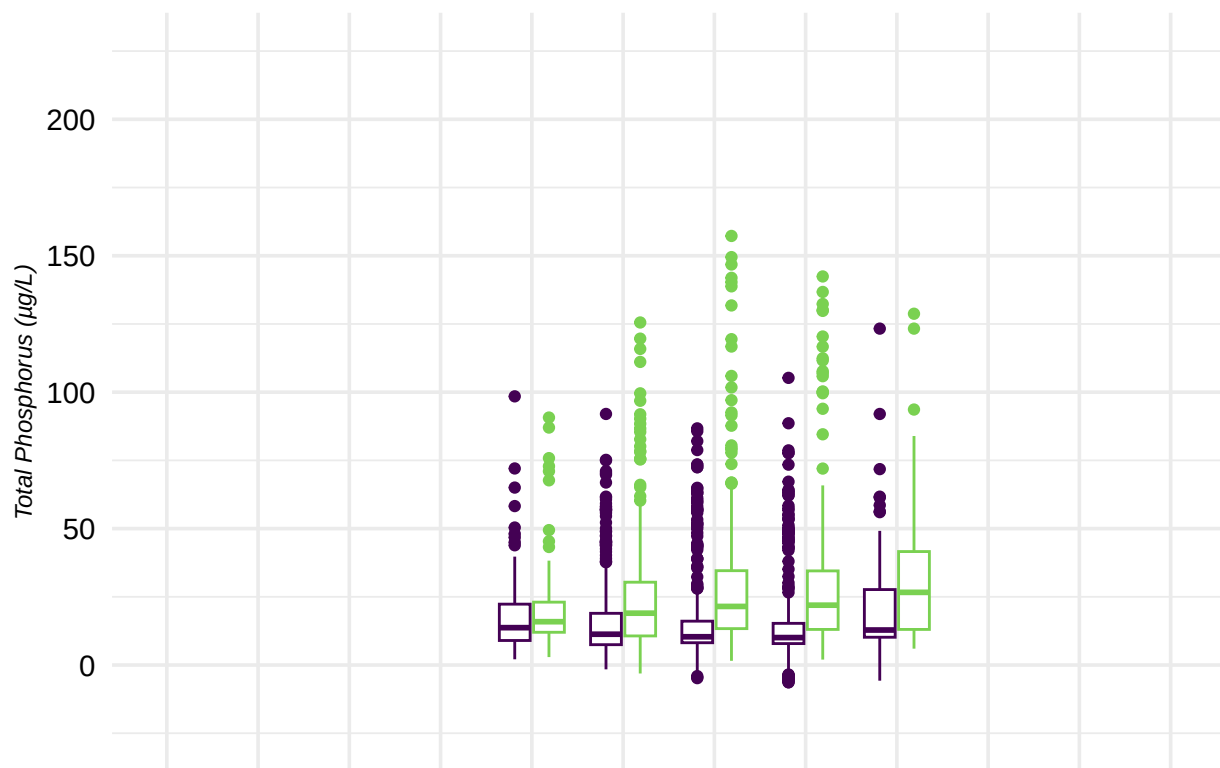
```

```

## Warning: Removed 20729 rows containing non-finite outside the scale range
## ('stat_boxplot()').

```

Monthly Phosphorus by Lake



#TN plot modified to remove legend and keep x-axis title and text

```

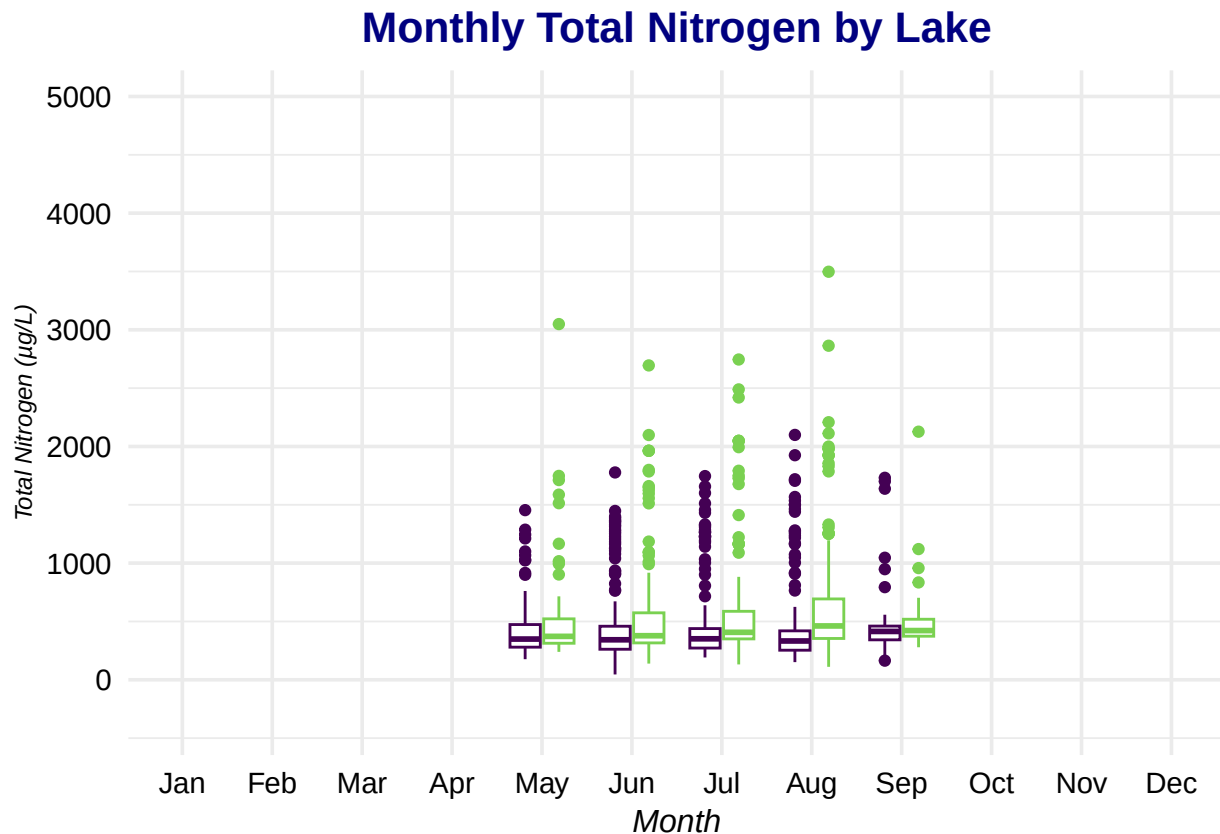
plot_tn_mod <- ggplot(
  PeterPaul.chem.nutrients,
  aes(
    x = factor(month, levels = 1:12, labels = month.abb),
    y = tn_ug,
    color = lakename)
) +
  geom_boxplot() +
  scale_color_viridis(discrete = TRUE, end = 0.8) +
  scale_x_discrete(drop = FALSE) +
  scale_y_continuous(expand = expansion(mult = c(0.2, 0.5))) +

```

```
labs(
  title = "Monthly Total Nitrogen by Lake",
  x = "Month",
  y = "Total Nitrogen (µg/L)",
  color = "Lake"
) +
mytheme +
theme(legend.position = "none",
      axis.title.y = element_text(face = "italic", size = 9))

print(plot_tn_mod)
```

```
## Warning: Removed 21583 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```



```
combined_cow <- plot_grid(
  plot_temp_mod,
  plot_tp_mod,
  plot_tn_mod,
  ncol = 1,
  align = "v",
  axis = "1")
```

```
## Warning: Removed 3566 rows containing non-finite outside the scale range
```

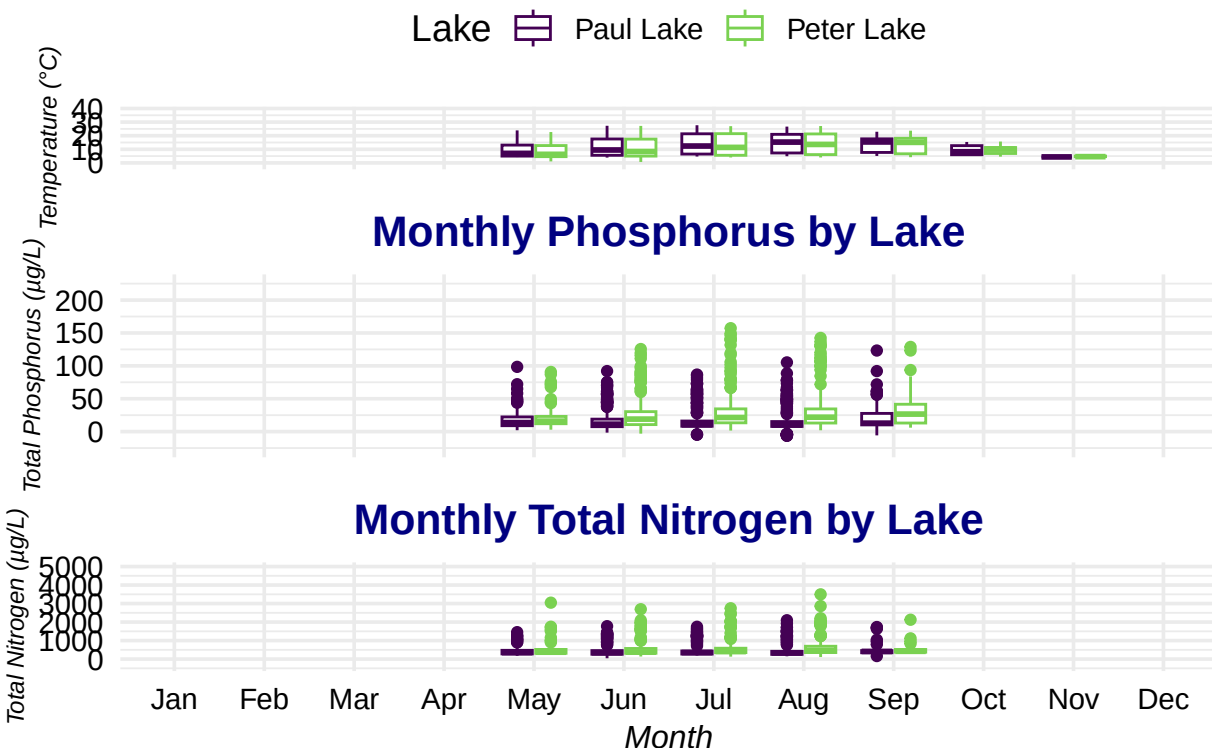
```
## ('stat_boxplot()').
```

```
## Warning: Removed 20729 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

```
## Warning: Removed 21583 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

```
print(combined_cow)
```

Monthly Temperature by Lake



Question: What do you observe about the variables of interest over seasons and between lakes?

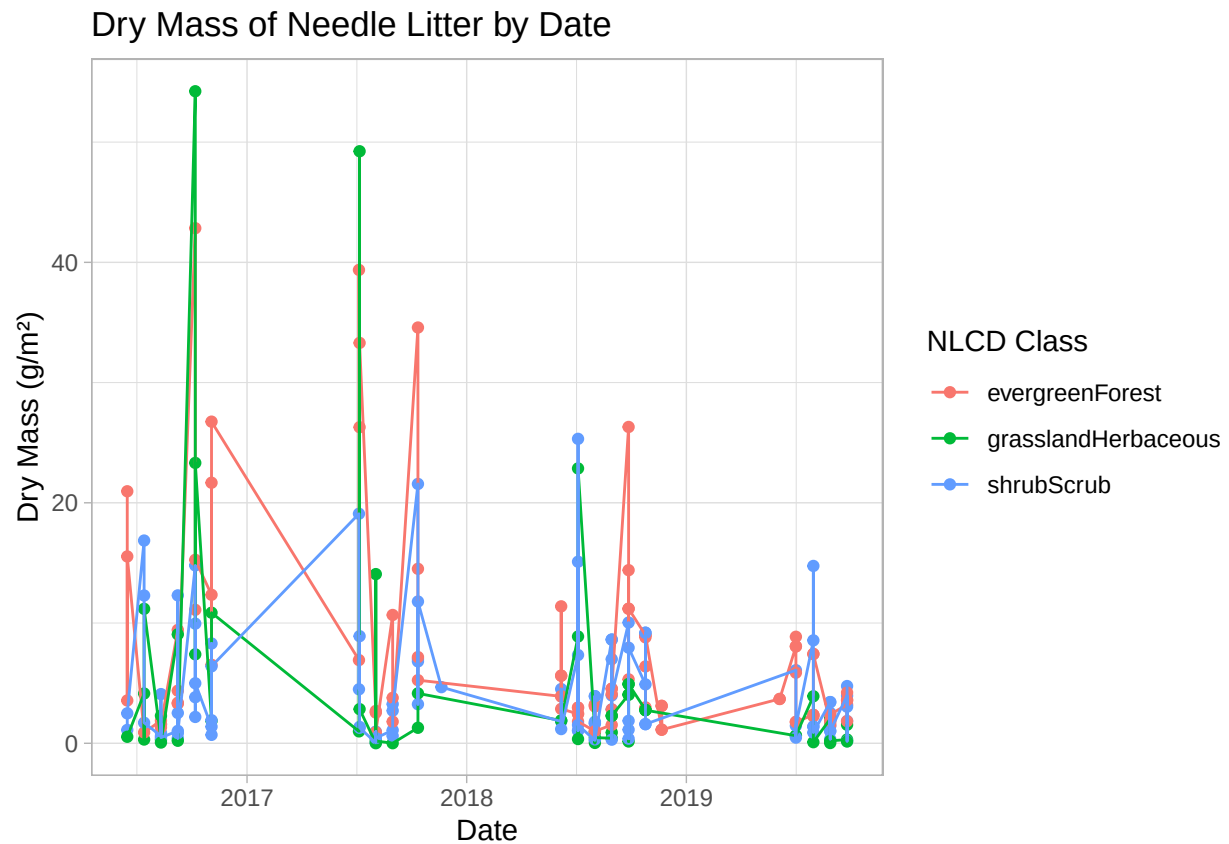
Answer: Temperature, phosphorus, and nitrogen all appear to be related. As lake temperature increases during the summer months, phosphorus and nitrogen levels also rise. Both nutrients reach their highest concentrations between June and August. Peter Lake consistently shows higher levels of phosphorus and nitrogen, which may indicate that it is more nutrient-rich or that it is experiencing eutrophication. It would be useful to investigate why Peter Lake has higher nutrient levels—for example, is there more runoff entering that lake?

6. [Niwot Ridge] Plot a subset of the litter dataset by displaying only the “Needles” functional group. Plot the dry mass of needle litter by date and separate by NLCD class with a color aesthetic. (no need to adjust the name of each land use)
7. [Niwot Ridge] Now, plot the same plot but with NLCD classes separated into three facets rather than separated by color.

#6

```
litter_needles <- NEON_NIWO_Litter %>%
  filter(functionalGroup == "Needles")

ggplot(
  litter_needles,
  aes(x = collectDate, y = dryMass, color = nlcdClass)
) +
  geom_point() +
  geom_line() +
  labs(
    title = "Dry Mass of Needle Litter by Date",
    x = "Date",
    y = "Dry Mass (g/m2)",
    color = "NLCD Class"
  ) +
  theme_light()
```

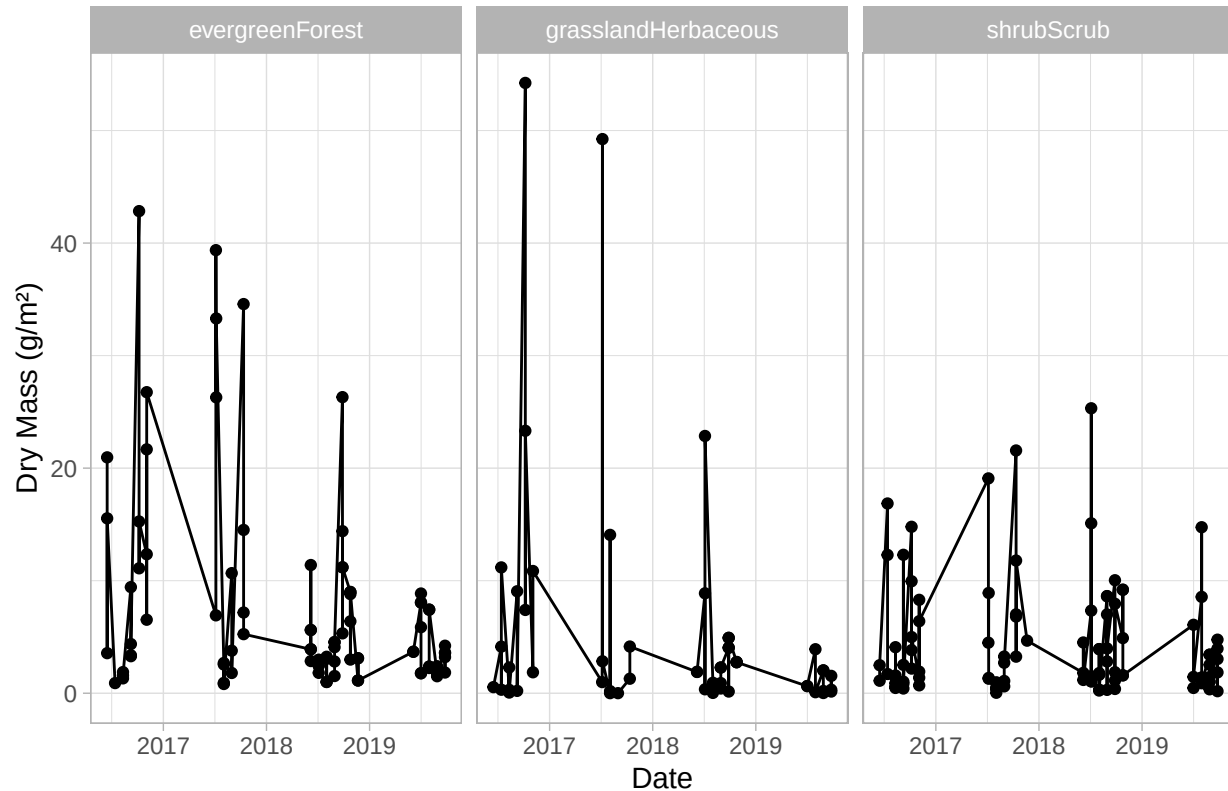


#7

```
ggplot(
  litter_needles,
  aes(x = collectDate, y = dryMass)
) +
  geom_point() +
```

```
geom_line() +
facet_wrap(~nlcdClass) +
labs(
  title = "Dry Mass of Needle Litter by Date by NLCD Class",
  x = "Date",
  y = "Dry Mass (g/m2)"
) +
theme_light()
```

Dry Mass of Needle Litter by Date by NLCD Class



Question: Which of these plots (6 vs. 7) do you think is more effective, and why?

Answer: Plot 7 is more effective because faceting the NLCD classes into separate panels makes patterns within each land-cover type easier to see. In Plot 6, overlapping colors can make it difficult to distinguish classes, whereas Plot 7 presents each NLCD class clearly in its own panel, improving readability and comparison.